

YAMAGATA, Japan

WMO: 475880

Lat: 38.2556N

Lon: 140.3461E

Elev: 504

StdP: 14.43

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.9	23.8	15.7	12.4	34.1	17.5	13.5	32.4	12.4	38.1	11.5	37.6	3.3	170	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.7	92.2	75.9	89.5	74.9	86.6	73.4	77.3	88.2	76.3	86.3	75.2	84.2	6.5	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	131.3	81.7	73.4	126.6	81.0	72.4	122.4	80.2	41.2	88.5	40.1	86.6	39.1	84.5	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.2	11.6	10.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			17.8	96.7	2.1	2.1	16.3	98.2	15.0	99.5	13.9	100.7	12.3	102.2
			16.8	79.4	2.0	1.0	15.4	80.1	14.2	80.7	13.1	81.2	11.7	81.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	31.8	32.9	39.5	50.8	62.0	69.2	76.2	78.3	70.1	58.4	46.4	36.7	(6)
(7)		DBStd	17.10	4.12	4.57	5.63	6.57	5.52	4.58	4.99	5.07	6.07	5.60	5.88	4.93	(7)
(8)		HDD50	2001	565	480	329	69	1	0	0	0	5	140	412	(8)	
(9)		HDD65	4957	1030	900	790	428	124	15	0	19	216	559	876	(9)	
(10)		CDD50	3631	0	0	3	92	372	575	813	876	603	265	31	1	(10)
(11)		CDD65	1114	0	0	0	1	30	139	349	412	172	11	0	0	(11)
(12)		CDH74	8462	0	0	0	33	313	822	2663	3575	1020	36	0	0	(12)
(13)		CDH80	2916	0	0	0	4	59	181	973	1416	280	3	0	0	(13)
(14)	Wind	WSAvg	3.8	3.6	3.7	4.3	4.6	4.4	3.9	3.7	3.8	3.5	3.4	3.5	3.6	(14)
(15)	Precipitation	PrecAvg	54.4	4.4	3.1	3.3	3.0	3.4	4.7	7.2	6.1	5.4	4.6	4.3	4.9	(15)
(16)		PrecMax	64.1	7.4	5.1	5.6	5.0	5.7	9.7	16.5	17.6	11.5	10.9	6.6	10.6	(16)
(17)		PrecMin	42.8	2.6	1.3	1.2	0.4	1.3	2.2	2.9	1.9	1.3	1.1	1.0	1.6	(17)
(18)		PrecStd	5.7	1.2	0.9	1.3	1.3	1.3	1.9	3.0	3.6	2.7	2.4	1.3	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.7	52.7	65.9	78.8	85.6	88.2	94.7	96.1	90.7	78.2	67.3	56.2	(19)
(20)		0.4%	MCWB	41.2	43.6	50.0	57.5	63.7	68.6	76.4	76.4	74.7	65.7	55.6	48.6	(20)
(21)		2%	DB	43.1	46.8	58.8	72.3	81.1	84.6	91.6	93.2	86.5	73.8	62.8	50.8	(21)
(22)		2%	MCWB	38.1	39.9	46.3	54.3	61.2	67.4	75.9	76.4	73.2	63.3	53.7	45.2	(22)
(23)		5%	DB	40.3	43.3	53.8	67.5	77.5	81.7	88.8	90.6	83.1	70.7	59.4	47.3	(23)
(24)		5%	MCWB	36.1	37.3	44.0	52.8	60.0	66.6	74.9	75.7	71.5	61.3	52.2	42.4	(24)
(25)		10%	DB	38.0	40.4	49.6	63.4	73.7	78.9	85.6	87.7	79.9	67.8	56.2	44.5	(25)
(26)		10%	MCWB	34.7	35.6	41.6	50.9	58.7	66.1	73.5	75.0	69.9	59.5	50.3	40.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.9	44.2	51.1	62.5	67.2	73.4	78.7	79.1	76.4	68.8	58.6	49.9	(27)
(28)		0.4%	MCDWB	46.6	50.2	61.7	68.9	76.9	81.2	90.0	90.6	86.2	74.6	63.4	54.4	(28)
(29)		2%	WB	38.9	40.7	47.9	57.5	64.7	71.1	77.2	77.7	74.6	65.6	55.5	45.8	(29)
(30)		2%	MCDWB	42.1	46.2	56.0	67.6	75.1	78.4	87.8	88.8	83.6	70.3	60.3	50.2	(30)
(31)		5%	WB	36.8	38.2	45.1	54.7	62.7	69.6	76.0	76.7	73.1	63.2	53.2	43.0	(31)
(32)		5%	MCDWB	39.4	42.1	51.6	64.1	72.0	77.2	85.7	87.3	80.9	68.3	57.6	46.5	(32)
(33)		10%	WB	35.0	36.1	42.6	52.5	61.0	68.0	74.8	75.8	71.5	61.0	50.9	40.7	(33)
(34)		10%	MCDWB	37.5	39.8	49.2	61.2	70.0	76.0	83.4	85.3	78.0	66.1	55.6	43.9	(34)
(35)	Mean Daily Temperature Range		MDBR	10.1	12.0	15.5	19.5	19.8	16.9	15.1	15.7	15.2	16.1	14.6	10.9	(35)
(36)		5% DB	MCDBR	13.6	16.9	24.1	27.2	27.3	22.2	19.9	19.7	18.6	18.7	19.7	16.0	(36)
(37)			MCWBR	9.4	11.1	14.0	13.2	11.4	8.9	7.3	6.7	7.9	9.6	11.9	11.1	(37)
(38)			MCDBR	13.1	16.3	21.6	22.4	20.9	16.9	17.9	17.8	16.8	15.6	17.7	15.5	(38)
(39)		5% WB	MCWBR	9.6	11.5	14.0	12.5	10.3	7.7	7.1	6.7	7.7	9.0	11.9	11.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.380	0.421	0.489	0.531	0.545	0.544	0.553	0.539	0.472	0.430	0.403	0.387	(40)	
(41)		taud	1.994	1.891	1.842	1.832	1.853	1.934	1.976	2.011	2.164	2.183	2.158	2.084	(41)	
(42)		Ebn at Noon	239	245	242	241	240	240	237	236	246	243	232	227	(42)	
(43)		Edh at Noon	41	52	61	65	65	60	57	54	43	39	35	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	551	798	1114	1428	1600	1549	1384	1410	1154	908	630	468	(44)	
(45)		RadStd	46	58	95	143	163	104	208	167	90	82	44	45	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.59	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page